

电路理论 mooc 答案——第十一章

仅限上海交通大学电路理论学科营使用

解析

请勿用于其他用途

1.解:

$$(1) f(t) = 1 - e^{-at} \xleftrightarrow{\mathcal{F}} F(t) = \frac{1}{s} - \frac{1}{s+a}$$

$$(2) f(t) = \sin(\omega t + \varphi) \epsilon(t) = (\cos \varphi \sin \omega t + \sin \varphi \cos \omega t) \epsilon(t) \xleftrightarrow{\mathcal{F}} F(t) = \frac{\omega \cos \varphi + s \sin \varphi}{s^2 + \omega^2}$$

Remark: $f(t) = \sin\left(\omega\left(t + \frac{\varphi}{\omega}\right)\right) \epsilon(t) \neq \sin\left(\omega\left(t + \frac{\varphi}{\omega}\right)\right) \epsilon\left(t + \frac{\varphi}{\omega}\right)$. 题目表述不严谨.

$$(3) f(t) = t + 2 + 3\delta(t) \xleftrightarrow{\mathcal{F}} F(t) = s^{-2} + 2s^{-1} + 3 = \frac{3s^2 + 2s + 1}{s^2}$$

$$(4) f(t) = e^{-at} + at - 1 \xleftrightarrow{\mathcal{F}} F(t) = \frac{1}{s+a} + as^{-2} - s^{-1} = \frac{a^2}{s^2(s+a)}$$

选A

2.解:

$$(1) F(t) = \frac{2}{(s+\alpha)^3} \xleftrightarrow{\mathcal{F}^{-1}} f(t) = t^2 e^{-\alpha t}$$

$$(2) F(t) = \frac{6s^2 + 26s + 26}{(s+1)(s+2)(s+3)} = \frac{3}{s+1} + \frac{2}{s+2} + \frac{1}{s+3} \xleftrightarrow{\mathcal{F}^{-1}} f(t) = (3e^{-t} + 2e^{-2t} + e^{-3t}) \epsilon(t)$$

$$(3) F(t) = \frac{7s^2 + 63s + 134}{(s+3)(s+4)(s+5)} = \frac{4}{s+3} + \frac{6}{s+4} - \frac{3}{s+5} \xleftrightarrow{\mathcal{F}^{-1}} f(t) = (4e^{-3t} + 6e^{-4t} - 3e^{-5t}) \epsilon(t)$$

$$(4) F(t) = \frac{5s^2 + 29s + 32}{(s+2)(s+4)} = 5 - \frac{3}{s+2} + \frac{2}{s+4} \xleftrightarrow{\mathcal{F}^{-1}} f(t) = (-3e^{-2t} + 2e^{-4t}) \epsilon(t) + 5\delta(t)$$

选D

3.解:

$$(1) F(t) = \frac{4s^2 + 7s + 1}{s(s+1)^2} \xleftrightarrow{\mathcal{F}^{-1}} f(t) = t^2 e^{-\alpha t}$$

$$(2) F(t) = \frac{2s^3 + 8s^2 + 2s - 4}{(s+1)(s+4)} = 2s - 2 + \frac{4}{s+4} \xleftrightarrow{\mathcal{F}^{-1}} f(t) = 2 \frac{d\delta(t)}{dt} - 2\delta(t) + 4e^{-4t} \epsilon(t)$$

$$(3) F(t) = \frac{20s^2 + 16s + 12}{(s+1)(s^2 + 2s + 5)} = \frac{4}{s+1} + \frac{8+6j}{s+1+2j} + \frac{8-6j}{s+1-2j} \xleftrightarrow{\mathcal{F}^{-1}} f(t) = (4e^{-t} + (8+6j)e^{-(1+2j)t} + (8-6j)e^{-(1-2j)t}) \epsilon(t) = 4e^{-t} \epsilon(t) (1 - 3\sin(2t) + 4\sin(2t))$$

$$(4) F(t) = \frac{100}{s^2(s+5)} = 20s^{-2} - 4s^{-1} + \frac{4}{s+5} \xleftrightarrow{\mathcal{F}^{-1}} f(t) = (20t - 4 + 4e^{-5t}) \epsilon(t)$$

选C

4.解:

$$Z = R_1 + (sL) // \left(\frac{1}{sC} + R_2 \right) = \frac{36s^2 + 26s + 2}{6s^2 + 12s + 1}$$

选D

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6.解:

先计算电感的初态电流 $i_L(0^-)$, $KCL: i_L(0^-) = -\left(C \frac{du_1}{dt} + \frac{u_1}{R_1}\right) \Big|_{t=0^-} = -\frac{3}{2} A$

对电容以外的电路做戴维南等效:

$$I_{SC} = \frac{1}{s} + \frac{2 \cdot \frac{3}{2}}{2+2s} = \frac{5s+2}{2s(s+1)}, Z_{eq} = R_1 // (R_2 + sL) = \frac{2(2+2s)}{4+2s} = \frac{2(s+1)}{s+2}, \text{ then } U_{OC} = \frac{5s+2}{s(s+2)}$$

$$KVL: U_1 + Z_{eq} \cdot C(sU_1 - u_1(0^-)) = U_{OC}$$

$$\text{Therefore, } U_1 = \frac{U_{OC} + Z_{eq} \cdot C u_1(0^-)}{1 + Z_{eq} \cdot sC} = \frac{\frac{5s+2}{s(s+2)} + \frac{2(s+1)}{s+2} \cdot \left(\frac{1}{2}\right)}{1 + \frac{2(s+1)}{s+2} \cdot \left(\frac{1}{2}s\right)} = \frac{4}{s^2 + 2s + 2} + \frac{1}{s} \Rightarrow u_1(t) = (1 + 4e^{-t} \sin t) \epsilon(t)$$

选A

7.解:

不妨设左回路 u_s 下侧和右回路 L_2 下侧均为零电势(每个回路都可以设个参考电平), 那么, 我们可以将 L_1 、 L_2 下侧相连, 进行 T 型去耦。

T 型去耦后直接求电路总输入阻抗, $Z_{eq} = R_1 + s(L_1 - M) + (sM) // [s(L_2 - M) + R_2] = 1 + 0.05s + \frac{0.05s \cdot (0.05s + 1)}{0.05s + (0.05s + 1)} = 1.25 + 0.075s - \frac{2.5}{s + 10}$

$$I_1 = \frac{U_s}{Z_{eq}} = \frac{\frac{2}{s}}{1.25 + 0.075s - \frac{2.5}{s + 10}} = \frac{1}{3} \frac{80(s + 10)}{s(s + \frac{20}{3})(s + 20)} = \frac{2}{s} - \frac{1}{s + 20} - \frac{1}{s + \frac{20}{3}} \Rightarrow i_1 = \left(1 - \frac{1}{2}e^{-\frac{20}{3}t} - \frac{1}{2}e^{-20t}\right)\epsilon(t)$$

$$sM, [s(L_2 - M) + R_2] \text{ 分流, } I_2 = -I_1 \cdot \frac{sM}{sM + [s(L_2 - M) + R_2]} \\ = -\frac{1}{3} \frac{80(s + 10)}{s(s + \frac{20}{3})(s + 20)} \cdot \frac{0.05s}{0.05s + (0.05s + 1)} = -\frac{40}{3} \frac{1}{(s + \frac{20}{3})(s + 20)} = -\frac{1}{s + \frac{20}{3}} + \frac{1}{s + 20} \Rightarrow i_2 = \left(-e^{-\frac{20}{3}t} + e^{-20t}\right)\epsilon(t)$$

选B

8.解:

$$t = 0^-, i_1(0^-) = \frac{u_s}{R_2} = 2A, i_2(0^-) = 0A.$$

该题计算简单, 过程略, 考虑 $t = 0^-$ 时 i_1 、 i_2 的取值, 选A.

9.解:

Complete response = zero-input response + zero-state response, i.e. 全响应 = 零输入响应和零状态响应之和

$$\begin{cases} 6 + \frac{2}{s+0.2} = \frac{6}{s} H(s) + U' \\ 9 - \frac{1}{s+0.2} = \frac{12}{s} H(s) + U' \end{cases} \Rightarrow \begin{cases} H(s) = \frac{s}{6} \left(3 - \frac{3}{s+0.2}\right) \\ U' = 3 + \frac{5}{s+0.2} \end{cases}$$

$$U_o(s) = \frac{s}{6} \left(3 - \frac{3}{s+0.2}\right) \cdot \frac{6}{s+5} + 3 + \frac{5}{s+0.2} = 3 + \frac{5.125}{s+0.2} - \frac{0.125}{s+5}$$

选D

10.解:

共有7个电感, 4个电容, 至多11个固有频率。

共有三个零阶线性微分方程约束, 所以 固有频率数 = $11 - 3 = 8$

$$\begin{cases} KVL: u_{C1} + u_{C2} + u_{C4} = 0 \\ KCL: i_{L1} + i_{L2} + i_{L6} = 0 \\ KCL: i_{L5} + i_{L7} + i_s = 0 \end{cases}$$

共有两个一阶线性微分方程约束, 所以 零固有频率数 = 2

$$\begin{cases} KVL: L \frac{di_{L2}}{dt} + L \frac{di_{L3}}{dt} + L \frac{di_{L4}}{dt} = 0 \\ KCL: C \frac{du_{C1}}{dt} + C \frac{du_{C3}}{dt} + C \frac{du_{C4}}{dt} = 0 \end{cases}$$

选C



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